

Evgenia-Maria S. Kontopoulou

CONTACT INFORMATION	High Performance Computing Lab Computer Engineering & Informatics University of Patras	e-mail: kontopoulo@upatras.gr webpage: https://eugeniamaria.github.io/ github: https://github.com/eugeniamaria
RESEARCH INTERESTS	(Randomized) Numerical Linear Algebra, Scientific Computing, High-Performance Computing, Big Data, Numerical Optimization, Information Retrieval	
EDUCATION	<i>Doctor of Philosophy (Ph.D.)</i> <i>Computer Science, Purdue University</i> <i>Advisor: Petros Drineas</i>	2016 - 2020
	<i>Bachelor of Engineering (B.Eng.) & Master in Engineering (M.Eng.)</i> Computer Engineering & Informatics, University of Patras <i>Advisor: Efstratios Gallopoulos</i>	2006 - 2012
WORKING EXPERIENCE	Independent Contractor, EasyHPC, University of Patras, Postdoctoral Researcher, University of Patras, Adjunct Lecturer, University of Patras, Senior Engineer, The MathWorks, Inc. Research Assistant, Purdue University Software Engineer Intern, The MathWorks Inc. Software Engineer Intern, The MathWorks Inc. Research Visitor, Rensselaer Polytechnic Institute Teaching Assistant, University of Patras Assistant for the Open Courses Project, University of Patras External Partner, Greek Research & Technology Network	November 2025 - May 2026 2025 - today Spring 2025 - today 2020 - 2024 2016 - 2020 May - August, 2019 May - November, 2018 2015 - 2016 2013 - 2015 2013 - 2015 2013 - 2014
PUBLICATIONS	G. Mermigkis, A. Sofotasios, E. Kontopoulou, E. Gallopoulos, P. Hadjidoukas, “ <i>DistPCA: Tera-Scale Genomic PCA via Out-of-Core Distributed Parallelism</i> ”, under review (2026) E. Kontopoulou, “ <i>Randomized Numerical Linear Algebra Approaches for Approximating Matrix Functions</i> ”, Ph.D. Dissertation (2020), Purdue University E. Kontopoulou, G. Dexter, W. Szpankowski, A. Grama, P. Drineas, “ <i>Randomized Linear Algebra Approaches to Estimate the Von Neumann Entropy of Density Matrices</i> ”, IEEE Transactions on Information Theory (2020), Vol. 66, Issue 8, pp. 5003-5021 A. Bose, V. Kalantzis, E. Kontopoulou, M. Elkady, P. Paschou, P. Drineas, “ <i>TeraPCA: a Fast and Scalable Software Package to Study Genetic Variation in Tera-scale Genotypes</i> ”, in Oxford Bioinformatics (2019), Issue 1, pp. 3679-3683 E. Kontopoulou, A. Grama, W. Szpankowski, P. Drineas, “ <i>Randomized Linear Algebra Approaches to Estimate the Von Neumann Entropy of Density Matrices</i> ”, in Proceedings of the 2018 IEEE International Symposium on Information Theory (2018), pp. 2486-2490	

P. Drineas, I. Ipsen, E. Kontopoulou, M. Magdon-Ismail, “*Structural Convergence Results for Low-Rank Approximations from Block Krylov Spaces*”, in SIAM Journal on Matrix Analysis and Applications (2018), Vol. 39, Issue 2, pp. 567-586

C. Boutsidis, P. Drineas, P. Kambadur, E. Kontopoulou, A. Zouzias, “*A Randomized Algorithm for Approximating the Log Determinant of a Symmetric Positive Definite Matrix*”, in Linear Algebra and its Applications (2017), Vol. 533, pp. 95-117

K. Fountoulakis, A. Kundu, E. Kontopoulou, P. Drineas, “*A Randomized Rounding Algorithm for Sparse PCA*”, in ACM Transactions on Knowledge Discovery from Data (2017), Vol. 11, Issue 3, No. 38, pp. 1-26

E. Kontopoulou, M. Predari, E. Gallopoulos, “*Onomatology and Content Analysis of Ergodic Literature*”, in Proceedings of the 3rd ACM Narrative and Hypertext Workshop (2013), No. 5, pp. 1-5

E. Kontopoulou, M. Predari, T. Kostakis, E. Gallopoulos, “*Graph and Matrix Metrics to Analyze Ergodic Literature for Children*”, in Proceedings of the 23rd ACM Conference on Hypertext and Social Media (2012), pp. 133-142

SELECTED CODE
BASED RESEARCH
PROJECTS

DistPCA
[GitHub Page](#)

C/C++, OpenMP, MPI

A distributed out-of-core C++ framework for tera-scale genomic Principal Component Analysis (PCA), designed to scale efficiently across both single- and multi-node computing systems. Built on top of Message Passing Interface (MPI), it employs a hybrid multi-level parallelism scheme combining multiprocessing, OpenMP multithreading, SIMD vectorization, and double buffering across all three stages of the PCA pipeline (I/O, data preprocessing, numerical method).

TeraPCA
[GitHub Page](#)

C/C++, OpenMP

A library that computes the top Principal Components (PCs) of tera-scale matrices using Randomized Singular Value Decomposition (RandSVD). Our implementation is based on multi-threaded libraries such as LAPACK, BLAS and MKL, and it can handle datasets that exceed the amount of available system memory by performing out-of-core computations.

Large Scale Genetic Data Simulator
[GitHub Page](#)

C/C++, OpenMP

A software that enables fast generation of simulated large scale genetic data using sophisticated random distributions to simulate genetic patterns.

Text-to-Matrix Generator Version 7.8

MATLAB

Version 7.8 introduces a new set of operations. More specifically we implemented deterministic and partially randomized algorithms for skeleton matrix decomposition. The skeleton matrix decomposition methods (e.g. CUR, CX, RRQR e.t.c.) utilize scaled parts of the initial matrix to derive highly accurate and interpretable approximations of the term-by-document matrix.

The highlights of version 6.7 are the incorporation of filters that enable parsing and preprocessing non-ASCII documents and additional options that rule the construction of the dictionary. Users can process a variety of data formats (e.g., pdf, docx, doc, ps e.t.c.) and select among many more options on how to construct the dictionary (e.g., exclude alphanumerics, numerics e.t.c.).

**AWARDS &
SCHOLARSHIPS**

SIAM Early Career Travel Award for SIAM LAA	April 2021
Purdue University Nomination for the Google Fellowship 2019 (one of the two nominees selected university-wise)	December 2018
John R. Rice Partial Fellowship in Scientific Computing	April 2018
Purdue CS Scholarship for Grace Hopper	April 2018
Gerondelis Foundation Scholarship	November 2017
NSF grant for XX Householder Symposium on Numerical Linear Algebra	June 2017
Scholarship for the 26th PCMI Summer Session, "The Mathematics of Data"	March 2016
Scholarship for the CRA-W Grad Cohort Workshop	January 2016
Scholarship for the 6th Gene Golub SIAM Summer School (selected as one of the 56 participants on international level)	June 2015
NAG Prize for the Best Presentation of the Session, "Matrix Algorithms and HPC for Large Scale Data Analysis", ERCIM	December 2013
SIGWEB Ted Nelson Newcomer Award, ACM Hypertext	June 2012
SIGWEB Student Travel Award for ACM Hypertext	March 2012

**SELECTED
TALKS**

"RandNLA Approaches to Estimate Logarithm-based Matrix Function", Invited Talk, 2021 SIAM Conference on Applied Linear Algebra and Applications (LAA21), May 2021

"Randomized Linear Algebra Approaches to Estimate the Von Neumann Entropy of Density Matrices", 2018 IEEE International Symposium on Information Theory (ISIT), Vail, CO, June 2018

"Towards Randomized Algorithms for the Estimation of Log-Determinants and Von-Neumann Entropies", Purdue Theoretical Computer Science Seminars, West Lafayette, IN, October 2017

"Structural Convergence Results for Low-Rank Approximations from Block Krylov Spaces", XX Householder Symposium, Blacksburg, VA, June 2017

"A Randomized Rounding Algorithm for Sparse PCA", Purdue Numerical Linear Algebra Student Seminar, West Lafayette, IN, April 2017

"Randomized Algorithm for Approximating the Log Determinant of a Symmetric Positive Definite Matrix", CSESC 2017, Purdue University, West Lafayette, IN, April 2017

"The Text-to-Matrix Generator", Sixth Gene Golub SIAM Summer School, Delphi, GR, June 2015

"Experiments with Randomized Algorithms in the Text to Matrix Generator Toolbox", ERCIM 2013, London, GB, December 2013

"TMG a MATLAB Tool for Text Mining", Numerical Linear Algebra Day, Athens, GR, November 2011

TEACHING	Numerical Analysis, UPatras (CEID NY240) Linear Algebra, UPatras (CEID NY110) Numerical Analysis, UPatras (CEID NY240)	Spring 2026 Fall 2025 Spring 2025
TEACHING ASSISTANCE	Foundations of Computer Science, RPI (CSCI 2200) Cryptography and Network Security, RPI (CSCI 4230) Numerical Analysis, UPatras (HY240) Linear Algebra, UPatras (HY110) Scientific Computing I, UPatras (HY343) Linear Algebra, UPatras (HY110)	Spring 2016 Fall 2015 Spring 2015 Spring 2014 Fall 2014 Spring 2013
ACADEMIC PEER-REVIEW	Grace Hopper Conference, Data Science Track 2018 International Symposium on Mathematical Foundations of Computer Science (MFCS) IEEE Transactions on Signal Processing ACM Transactions on Mathematical Software (TOMS) SIAM Journal on Imaging Science (SIIMS) SIAM Journal on Matrix Analysis and Applications (SIMAX) SIAM Journal on Scientific Computing (SISC) 34th International Conference on Machine Learning (ICML)	2017 - 2019 2017 - today 2017 - today
QUALIFICATIONS & INFORMATION	<i>Programming Languages</i> MATLAB, C++, Python <i>Languages</i> Greek (native), English (proficient), German (intermediate) <i>Memberships</i> SIAM, ACM	
OTHER ACTIVITIES	Mentoring undergraduate students on their theses on the Text-to-Matrix Generator at University of Patras, Greece President at the Hellenic Student Association at Purdue University. Vice president at the Hellenic Student Association at Purdue University.	2019 - 2020 2017 - 2019
LAST UPDATE	July 12, 2026	